

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MTH 224 — Past Questions & Materials
by Renance

Course: MTH 224
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 200

This study pack was compiled and restructured by Renance for MTH 224 students at Federal University of Petroleum Resources, Effurun. It brings together 1 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

Study online: renance.app
Get the app: Google Play — [dev.renance.renance](https://play.google.com/store/apps/details?id=dev.renance.renance)

Generated by Renance AI

Paper 1 of 1 — MTH 224 2022/2023

Level 200 • Second Semester • PastQuestion

DEPARTMENT. OF MATHEMATICS

COLLEGE OF SCIENCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

SECOND SEMESTER B.Sc. EXAMINATIONS 2022/2023 SESSION

Course: MTH224 (Complex Analysis 1)

Instructions: Attempt any FOUR questions

hrs

Time allowed: 2%

Question One:

. where Z is the

(a) Define the limit of a function of a complex variable and prove that a
dz

conjugate of z . does not exist anywhere.

(b) , (i) Determine the locus represented by $|z-3| + |z+3|=10$

(ii) Express $f=$ in the form $u(x,y) + iv(y)$

State and prove Cauchy-Riemann equations

(c)

Question Two:

uchy's Theorem and verify the theorem by evaluating the integral $\oint_C f(z)dz$

'ound the square formed by joining the points $z=1, z=2, z=24, z=1$

for n positive integer

ide and on a simple closed curve ϕ , and a is any point within ϕ . prove

a

(ii) $\backslash z = -$

$> eee$.

= (iii) the rectangle whose vertices are

FEDERAL UNIVERSITY

OF PETROLEUM RESOU

ote ag ee eae Ee

f the parabola $y = 4-x^2$ from $A(2,0)$ to $B(-2,0)$

dz e cis that part 0

(b) / Evaluate , wher

LA +1

Question Four: oe ;

- (a) Define conformal mapping and state the Conditions for a transformation to be conformal.
- (b) Map the points $A(z = -2+i)$ and $B(z = 3/+4i)$ onto the w -plane under the transformation $w = 2iz + 3$ and illustrate the transformation on a diagram.
- Let w as z moves along the straight line joining $A(2+0i)$
- (c) If $w = 7$, find the path traced $0 \leq t \leq 1$ and $B(0+2i)$. Illustrate the transformation on a diagram.

4

onto the w -plane is conformal or not.

Show whether the transformation w

series in z , determine the radius of convergence

: $w = 30$

vergence interval (i) (ii) =

. $22-1 (2-22 +3)$

the following as po

a

jes expansion of $f(z) = \cos z$ about the point $z = -$

3

na Laurent series about the point $z = 2$ and determine the nature of the

yr and Laurent series

- for (i) $|z| < 2$ (ii) $|z| > 2$

(ii) $1 < |z| < 4$ (a) $|z| > 4$. In each case specify